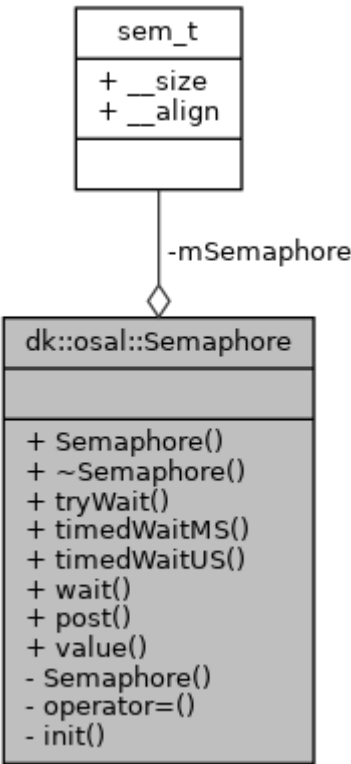


dk::osal::Semaphore Class Reference

```
#include <DKOsaiSemaphore.h>
```

Collaboration diagram for dk::osal::Semaphore:



Public Member Functions

	Semaphore (uint32_t counter)
	Constructor. More...
	~Semaphore ()
	Destructor. More...
bool	tryWait ()
	Try to wait for a semaphore If the counter is greater than "0", it will be decremented by "1". More...
bool	timedWaitMS (uint32_t msec)
	Blocks the current thread until the semaphore counter is greater than "0", and then decrements the counter by "1". If the specified timeout interval elapses, the function is cancelled. More...
bool	timedWaitUS (uint32_t usec)
	Semaphore wait operation (waits with a timeout) Blocks the current thread until the semaphore counter is greater than "0", and then decrements the counter by "1". If the specified timeout interval elapses, the function is cancelled. More...
void	wait ()
	Semaphore wait operation Blocks the current thread until the semaphore is greater than "0", and then decrements the counter by "1". More...

void **post** ()

Semaphore post operation Increments the semaphore counter by "1" and then notifies the waiting threads. [More...](#)

uint32_t **value** ()

Get counter value of semaphore. [More...](#)

Private Member Functions

Semaphore (const **Semaphore** &)

Disable Copy Constructor and Assignment operators. [More...](#)

Semaphore & **operator=** (const **Semaphore** &)

void **init** (uint32_t **counter**)

Initialize the **Semaphore**. [More...](#)

Private Attributes

sem_t mSemaphore

Constructor & Destructor Documentation

◆ Semaphore() [1/2]

dk::osal::Semaphore::Semaphore (uint32_t **counter**)

Constructor.

Parameters

counter Initial value of the semaphore counter.

requirement ID 751538,751584

◆ ~Semaphore()

dk::osal::Semaphore::~Semaphore ()

Destructor.

Parameters

counter Initial value of the semaphore counter.

requirement ID 751538,751584

◆ Semaphore() [2 / 2]

dk::osal::Semaphore::Semaphore (const **Semaphore** &)

private

Disable Copy Constructor and Assignment operators.

Member Function Documentation

◆ init()

void dk::osal::Semaphore::init (uint32_t **counter**)

private

Initialize the **Semaphore**.

requirement ID 751591

◆ operator=()

Semaphore& dk::osal::Semaphore::operator= (const **Semaphore** &)

private

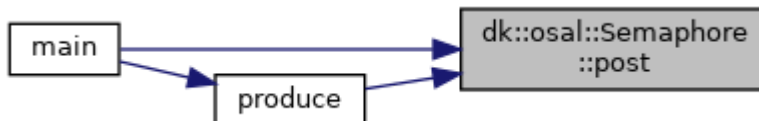
◆ post()

```
void dk::osal::Semaphore::post ( )
```

Semaphore post operation Increments the semaphore counter by "1" and then notifies the waiting threads.

requirement ID 751589

Here is the caller graph for this function:



◆ timedWaitMS()

```
bool dk::osal::Semaphore::timedWaitMS ( uint32_t msec )
```

Blocks the current thread until the semaphore counter is greater than "0", and then decrements the counter by "1". If the specified timeout interval elapses, the function is cancelled.

Parameters

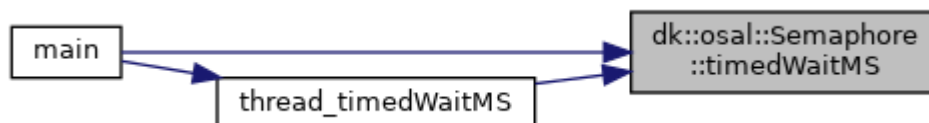
msec number of milliseconds to wait before the `wait()` is cancelled

Returns

true semaphore counter is greater than "0" **false** a timeout occurred

requirement ID 751586

Here is the caller graph for this function:



◆ timedWaitUS()

```
bool dk::osal::Semaphore::timedWaitUS ( uint32_t usec )
```

Semaphore wait operation (waits with a timeout) Blocks the current thread until the semaphore counter is greater than "0", and then decrements the counter by "1". If the specified timeout interval elapses, the function is cancelled.

Parameters

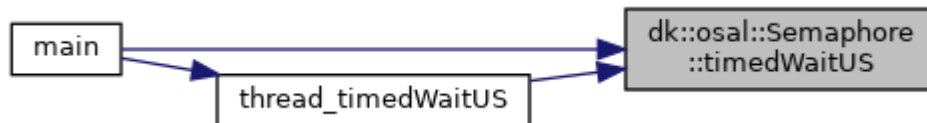
msec number of microseconds to wait before the **wait()** is cancelled

Returns

true semaphore counter is greater than "0" **false** a timeout occurred

requirement ID 751587

Here is the caller graph for this function:



◆ tryWait()

```
bool dk::osal::Semaphore::tryWait ( )
```

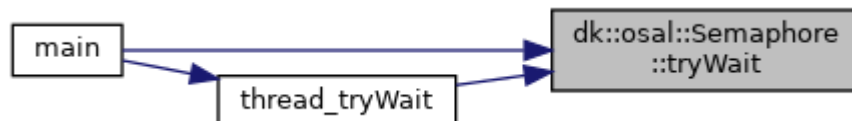
Try to wait for a semaphore If the counter is greater than "0", it will be decremented by "1".

Returns

true semaphore counter greater than "0" **false** semaphore counter is "0"

requirement ID 751585

Here is the caller graph for this function:



◆ value()

```
uint32_t dk::osal::Semaphore::value ( )
```

Get counter value of semaphore.

Returns

current counter value

requirement ID 751590

Here is the caller graph for this function:



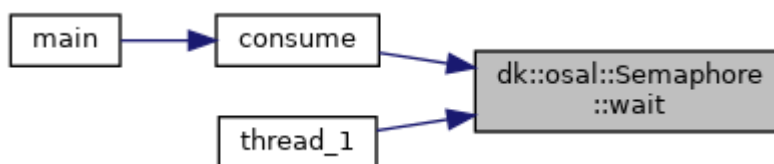
◆ wait()

```
void dk::osal::Semaphore::wait ( )
```

Semaphore wait operation Blocks the current thread until the semaphore is greater than "0", and then decrements the counter by "1".

requirement ID 751588

Here is the caller graph for this function:



Member Data Documentation

◆ mSemaphore

```
sem_t dk::osal::Semaphore::mSemaphore
```

private

The documentation for this class was generated from the following files:

- /home/damudhar/29_8_audi/cluster-platform/dijkstra/abstraction/osal/src/posix/**DKOsalSemaphore.h**
- /home/damudhar/29_8_audi/cluster-platform/dijkstra/abstraction/osal/src/posix/**DKOsalSemaphore.cpp**